

# End to End Evaluation of Fabric-X Committer and Orderer

Every comparison in this section is against the results reported in the SIGMOD’26 paper: its Figure 9 for the committer sweeps and its Figure 7b for throughput against transaction size. That is a different deployment from this one, so a difference between the two is a difference between deployments.

## Part I — The committer on its own

### 1 Setup: the committer machines

Nineteen machines, each two sockets of sixteen Intel Xeon Sapphire Rapids cores at 2.10 GHz — 64 hardware threads, 16 MiB of L3 per socket — with 156 GiB of RAM, running three signature verifiers, one coordinator, one sidecar, one load generator, six validator–committers, twelve YugabyteDB tablet servers and three YugabyteDB masters. The six validator–committers and the three masters sit on nine of the twelve database machines, and no machine runs both a validator–committer and a master.

The coordinator tracks dependencies with the simple manager — one map of keys to waiting transactions, owned by a single goroutine — rather than the sharded graph guarded by a mutex. Every machine on both arms runs Red Hat Enterprise Linux 9.8, kernel 5.14. Storage is two 969 GiB XFS volumes per machine, `noatime`, mounted `/data1` and `/data2`. Each database node passes YugabyteDB both volumes as separate data directories (`--fs.data.dirs`, one path per volume) rather than a stripe. The sidecar’s ledger uses `/data1` alone.

Every transaction slot takes a fresh key, so there are no read–write conflicts except in the panel that introduces them deliberately. There a share of transactions carries one back-reference: an existing key in a read–write slot with a nil expected version, which read validation rejects on the same path a double spend takes. That share is the panel’s x axis, and the reference is drawn from the newest million keys as they stood 300,000 transactions earlier. Blocks come from the load generator’s embedded mock ordering service, which cuts and signs them itself and serves them to the sidecar over the Atomic Broadcast API, so no ordering service is in the path.

Transaction signatures satisfy namespace 0’s endorsement policy, a threshold rule over the endorsing user’s signing certificate. The load generator creates the namespace itself on this arm, registering the policy over a key pair it holds, so the policy and the key it signs with agree by construction and the scheme is a free choice. These figures sign with ECDSA P-256, as Part II’s do.

A measured point is the highest offered rate that arrived in full, committed within 2%, kept p99 under 1s and held a flat in-flight count, confirmed by a 300s hold on a fresh deployment.

### 2 Why this setup

**The shape is not the SIGMOD’26 paper’s, and neither is the hardware.** Its committer deployment is nine validator–committers on nine database nodes, and both of its arms ran on AWS: a `c6id.8xlarge` for each validator–committer with its database node (32 vCPUs of Ice Lake at 3.5 GHz, 64 GiB), a `c6id.16xlarge` each for the coordinator, the sidecar and the load generator, and a `c5a.8xlarge` for each verifier (32 vCPUs of EPYC at 3.3 GHz). Storage on both arms is EBS General Purpose, rated at 3,000 IOPS and 125 MiB s<sup>-1</sup> — the `c6id` local NVMe was not used. This deployment is wider and slower per core — 64 hardware threads at 2.10 GHz a machine — with more memory, and with local volumes that Table 2 measures at two orders of magnitude more random write IOPS than that EBS rating. What follows therefore compares deployments, not matched hardware.

**Both volumes, not a stripe**, so YugabyteDB sees the disk boundary and places data across it itself.

**A conflict-free workload** measures the pipeline’s ceiling rather than its conflict behaviour, at the cost of a state database and a ledger that grow with every transaction.

**The reference gap is derived, not copied.** A back-reference is only a double spend if the key it names is already committed; a reference to a key still in flight is a dependency the coordinator has to order, which is a different measurement wearing the same label. How far back a reference must reach to be safe is a property of the deployment, not of the workload: in-flight keys are the rate times the latency times the fresh keys a transaction introduces, so 304,000 to 402,000 keys here at 518,000 tps and a 345 ms to 457 ms pipeline. A gap of 300,000 transactions is 510,000 keys and clears all of it. The SIGMOD’26 gap of 1,000 transactions clears its own 85 ms pipeline and would leave two fifths of these references inside this one, which is why the parameter is derived here rather than copied. Erring long costs nothing — an older key is still a double spend. Conflicts are then introduced deliberately as their own variable.

**A flat in-flight count is required** because throughput alone cannot tell a rate the pipeline sustains from one it is absorbing into a growing backlog: both commit everything offered for as long as the queue has room to grow. What separates them is latency, since a transaction waits in that queue before anything processes it, and what happens next — the queue is bounded, so a rate above capacity collapses once it fills.

### 3 Throughput and tail latency

Throughput falls with transaction size (Fig. 1a), 653,273 tps at one read–write operation to 274,727 at four. Comparing that with the SIGMOD’26 figure needs the axes aligned: its x axis counts inputs and outputs, and one in with one out is a *two* read–write total, so its *N/N* sits at our *2N*. Only two of our four sizes therefore have a published counterpart.

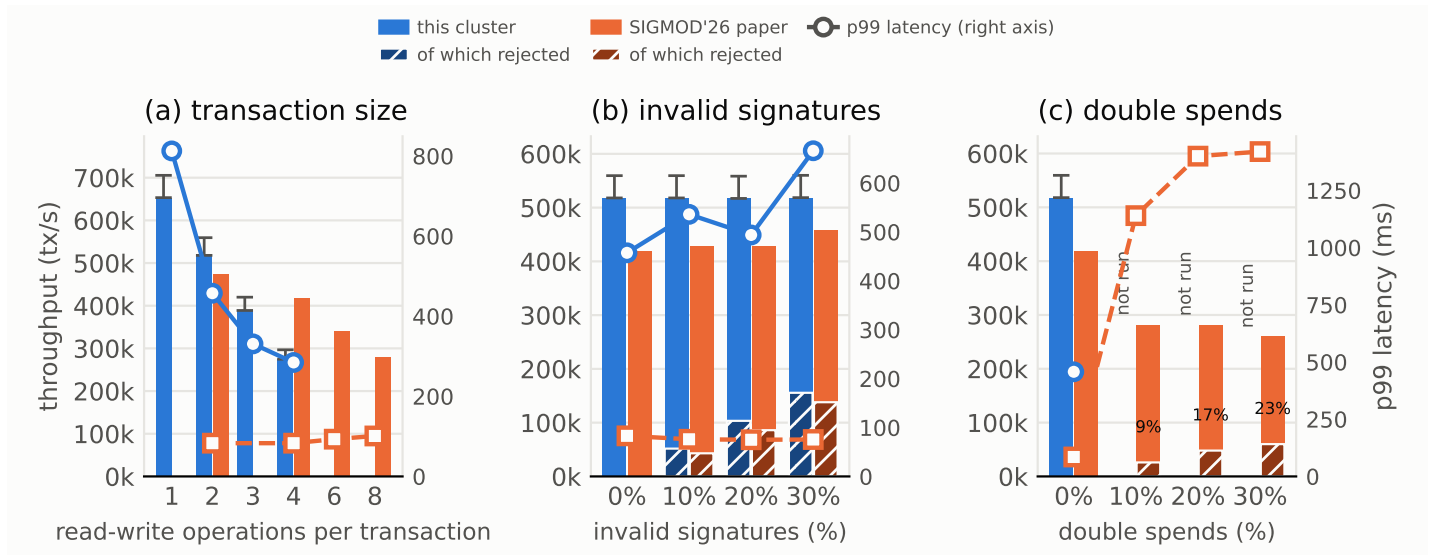


Figure 1: Committer throughput against (a) transaction size, (b) invalid-signature share and (c) double-spend share, beside the SIGMOD’26 paper’s Figure 9. Bars are throughput on the left axis; the marks above them are the 99th-percentile latency of the same measurement on the right axis, in that series’ own colour. Throughput counts transactions finished, committed plus rejected, and the hatched part of a bar is the rejected share — labelled with the share it measured only in (c), where what the generator produced differs from what was asked for; elsewhere it is the x value. Whiskers are the search’s 8% step, one-sided. *Not run* marks a condition never attempted at that configuration.

At two read-writes this cluster holds 518,000 tps against the SIGMOD’26 474,000, 9% ahead; at four it holds 274,727 against 419,000, 34% behind. Over those same two sizes our throughput falls 47% where the SIGMOD’26 sweep falls 12%, so whatever this pipeline pays per key, it pays much more steeply. Its six- and eight-key points were never measured here, and our one- and three-key points have nothing to compare against.

The mechanism behind that slope is not the SIGMOD’26 one, which is lock contention in the coordinator’s dependency graph. There is no such lock here: the simple manager has one owning goroutine where the sharded graph has a mutex, and its two lock-wait histograms read zero in every one of 2,326 thirty-second samples across these runs. What the graph costs instead is per key: a stage benchmark of the manager puts it at about 420ns a key, flat from one key per transaction to eight, which is a fall of the shape this panel shows. Running the sharded graph on this deployment instead does register on those histograms, so the difference is the manager and not the instrument.

Read as a key rate rather than a transaction rate, the difference between the two sweeps has a shape. Multiplying each point by its keys per transaction gives 653,273, 1,036,000, 1,166,727 and 1,098,908 keys a second at one to four keys: it climbs to about 1,170,000 and stops. Above one key this pipeline is therefore bounded per key, and its transaction rate falls because each transaction spends more keys out of a fixed key budget. The SIGMOD’26 sweep does not reach a boundary over its range — 948,000 keys a second at two keys, then 1,676,000, 2,040,000 and 2,240,000 at four, six and eight. At two keys our key rate is the higher of the two; by four theirs is 53% above ours and still rising.

Where that key budget goes is a hypothesis rather than a result, and the storage path is not the answer. Every key

costs a version lookup and a write in the state database, but the published deployment’s volumes are rated at 3,000 IOPS and  $125 \text{ MiB s}^{-1}$  against the 102,000 random-write IOPS and  $1,058 \text{ MiB s}^{-1}$  Table 2 measures here, so whatever the key budget is spent on, it is not disk this deployment lacks. Neither ceiling is disk-bound at these rates in any case: nine volumes at that rating supply 27,000 IOPS against the 1,676,000 keys a second the published sweep reaches at four keys, so both deployments must serve reads from memory and batch their writes.

That leaves the compute side, where the two differ in opposite directions. Each published validator-committer runs on 32 vCPUs at 3.5 GHz, 67% more clock per core than here; this deployment answers with more cores and more memory per machine, and with six validator-committers against nine. So the published pipeline is the faster one per key and this one the wider one, which is consistent with a per-key cost that is partly serialized — and that is as far as a ladder on one deployment can place it. Settling it needs the two tier sizes measured at a matched clock, which nothing here provides.

Invalid signatures are close to free (Fig. 1b): 518,000 tps with none, then 518,000, 517,091 and 518,545 at 10%, 20% and 30% — all four flat within 0.3%, far inside this cluster’s 8.8% repeatability, so rejection costs nothing measurable in throughput. It is not the SIGMOD’26 10% gain either.

Double spends are not free here, and what they cost is not conflict resolution. At the tablet layout this deployment uses, no offered rate between 25,000 and 500,000 tps qualified at a 5% double-spend share: every rung delivered 21,000 to 24,000 tps with a mean latency near 150s, so the figure is flat rather than sloped and the pipeline’s conflict ceiling is around 22,000 tps — a factor of twenty-four below the 518,000 it sustains with none.

The cost is a database read amplification, and three measurements place it there rather than in conflict handling. Per transaction, the busiest machine spends 2,133  $\mu$ s of CPU under conflicts against 99  $\mu$ s without them, so the work is real rather than queued — no queue makes a transaction cost twenty times the CPU. The insert step of a commit takes 1.72s a call at 1.91 calls a commit, the second call being a retry. And the same workload at eight tablets instead of 120 reaches 181,091 tps at 239 ms, with per-transaction CPU back to 75  $\mu$ s: the penalty is gone, not reduced.<sup>1</sup>

The mechanism is visible in the state database's own procedure. A batch of new keys is inserted blind and returns on success, performing no lookup at all. One key that already exists raises a uniqueness violation, and the handler then looks up *every* key the batch carries to report which ones collided — a multi-key lookup that YugabyteDB batches per tablet only while tablets times keys stays under about 32,768, and serves with one storage read per key above it. The batch is rolled back and retried, which is the second insert call. So a conflict-free workload never performs the lookup, one conflicting key makes a whole batch perform it, and whether that lookup is batched depends on a product this deployment sits on the wrong side of: about 304 keys an insert against 120 tablets.

That makes the conflict figure a property of the deployment's storage layout rather than of its conflict handling, which is why it is reported here rather than beside the published double-spend sweep: the published deployment's numbers were taken at a layout this one does not share.

## 4 Latency against throughput

Both curves carry two read-write operations over 32 B keys, 262 B serialized, and differ only in the number of transactions the generator puts in a block.

At 10,000-transaction blocks the median has a floor near 125 ms that does not fall with the rate, since nothing in a block moves until the block is cut (Fig. 2). At 500 transactions a block the floor is 40 ms, and 379,764 tps holds at 136 ms median and 235 ms p99 — about the median large blocks give at 10,000 tps, at thirty-eight times the rate.

That ladder stops where the load generator stops, not where the committer does, and the constraint is the interface between them rather than any cost inside either. What bounds the transactions the generator can have outstanding is a buffer counted in *blocks*, so at 500 transactions a block it holds twenty times fewer transactions than at 10,000 for the same rate. Deepening it twentyfold, to match in transactions, moves the small-block ceiling from 396,000 to 400,000 to 430,145 tps sustained at a 209 ms median, where the generator plateaus at 21% of its cores. What the committer itself sustains at this block size is therefore still not resolved: every rung measured so far is a property of the instrument.

<sup>1</sup>That rate is a 90s probe. Its 300s hold delivered about 131,000 tps with a 26s mean, so eight tablets raises the ceiling by an order of magnitude without making the conflict path healthy.

## Part II — End to end, with a real ordering service

The second arm puts a real Arma ordering service in the path. The committer tier is **unchanged from Part I** — the same nineteen machines, the same components, the same tuning — so a difference between the two arms is attributable to ordering.

## 5 Setup: the ordering machines

Twenty further machines of the same processor, each two sockets of eight cores — 32 hardware threads, 16 MiB of L3 per socket — with 78 GiB of RAM, holding four parties of a router, a consenter, an assembler and one batcher per shard. Routers, consenters and assemblers take a machine each, twelve in all; the batchers share the remaining eight. Two configurations are compared:

- **Four shards.** Sixteen batchers, two to a machine, each with a volume to itself (`/data1` and `/data2`).
- **Eight shards.** Thirty-two batchers, four to a machine, two sharing each volume.

Storage is two 485 GiB XFS volumes per machine, mounted as on the committer machines. Routers, consenters and assemblers write to `/data1`; the batchers are distributed across both volumes as above.

A batcher cuts batches and the assemblers turn each one into a block, so on this arm the block-size knob is the shared configuration's `Batching.BatchSize.MaxMessageCount` rather than a load-generator setting, and `BatchCreationTimeout` is 500 ms. The consensus decision interval, `RequestBatchMaxInterval`, is 50 ms, reduced from the default 500 ms.

## 6 Why this setup

**Two batchers to a machine** is measured rather than assumed. At two shards the service capped near 158,000 tps *per shard* whatever the batch size or decision interval, while batcher processes sat at 15% to 20% of a 32-core machine — so the constraint is per shard and the spare capacity for a second was already on the box.

**Both shard counts are measured** because that reading predicts more shards buy throughput, and the two configurations separate the two things a shard needs: a core budget and a disk. Four shards give each co-located batcher its own volume; eight give twice the shards but make two share one.

**These machines are half a committer machine**, and half is the whole of the difference. They are the same processor at the same clock, so a comparison between the arms is one of width rather than of generation. What differs is half the threads, half the memory and, per Table 2, half the disk bandwidth; what favours the ordering machines is twice the L3 per core. So a ceiling measured on this arm belongs to these machines before it belongs to ordering, while a per-core one belongs to neither class in particular.

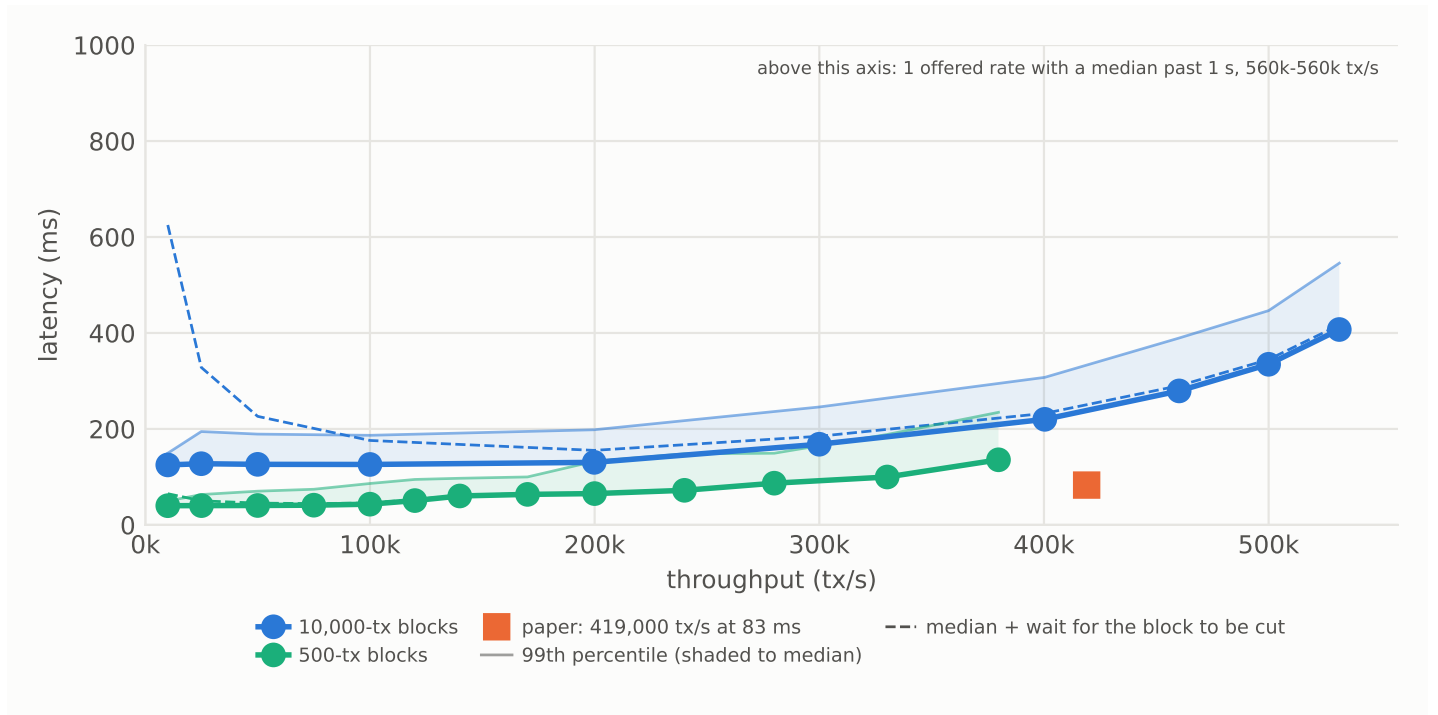


Figure 2: Latency against throughput for the committer alone, at two block sizes and a fixed workload of two read–write operations. Thick line: the median; the thin line above it the 99th percentile, shaded between; dashed: the median plus the mean wait for a block to be cut, which the measurement excludes. The latency axis runs to one second, the bound every reported point meets: it is floored so the operating region stays legible and capped so a rung that took seconds cannot flatten the rest. Offered rates whose median is past it are summarised in a corner rather than drawn, and a hollow marker appears only where one falls inside the axis.

A 50 ms **decision interval** because the default 500 ms was the largest single term in end-to-end latency and a ceiling on throughput as well. A consenter sees batch attestations rather than transactions — some 28 a second across two shards at 280,000 tps — so `RequestBatchMaxCount` of 100 would take seconds to fill and the interval is what cuts every decision. Throughput follows the interval for the same reason: at the ceiling each decision carries about the same number of transactions, so more decisions a second is more transactions a second.

## 7 Latency against throughput, by shard count and storage

Both ladders here carry the workload of Fig. 2 — two read–write operations over 32 B keys, 262 B serialized, every slot taking a fresh key — so the two arms are comparable at matched rates and differ only in whether an ordering service is in the path.

End to end, ordering included, this deployment sustains 499,091 tps at 658 ms p99, with every offered rate up to that arriving in full and no aborted transactions at any rung. Two comparisons place it. The SIGMOD’26 paper has no end-to-end figure, but its ordering service alone reaches 430,000 tps at four parties and four shards — so the complete pipeline here exceeds the published figure for one part of it. That is the one comparison in this section between comparable machines: those experiments ran on 32-vCPU VMs with 64 GiB of memory, against the 32 threads and 78 GiB here, and on

EBS volumes rated below what Table 2 measures on these. And the same committer at the same workload without an ordering service holds 518,000 tps, reaching 531,455 on the ladder of Fig. 2, so putting real ordering in the path costs about 6% of the committer’s own ceiling — inside this cluster’s repeatability rather than a separate order of magnitude.

Both configurations sustain the same throughput and hit the same knee: 499,091 tps at four shards against 500,000 at eight, 0.2% apart, and neither sustains 550,000 (Fig. 3). Four shards is the better configuration on every other axis — p99 lower at every matched rate, by 44 ms to 248 ms (658 ms against 754 ms at the knee), fewer processes, and a volume per batcher rather than two sharing one.

Doubling the shards buys nothing because ordering is not what limits this arm. Below 400,000 tps the busiest machine is a router; at the top of both ladders it is a validator–committer at 76% to 82%. Extra shards add capacity to the tier that is not binding.

The knee itself cannot be attributed to a single tier. At 500,000 tps no machine anywhere exceeds 82%: the busiest validator–committer is at 81% and *the load generator is at 77%*, four points behind it. So the ceiling is neither a CPU wall on any tier nor separable by this experiment — the two candidates are within each other’s margin, and a ladder on one deployment cannot tell them apart. Naming the busiest host names a correlate, not a cause.

Extra shards also cost latency, for a reason the block size makes plain. At a given total rate each of eight shards sees half the per-shard rate that four do, so batches take twice as long to reach 10,000 transactions and the 500 ms `BatchCreationTimeout` cuts them early over twice as wide

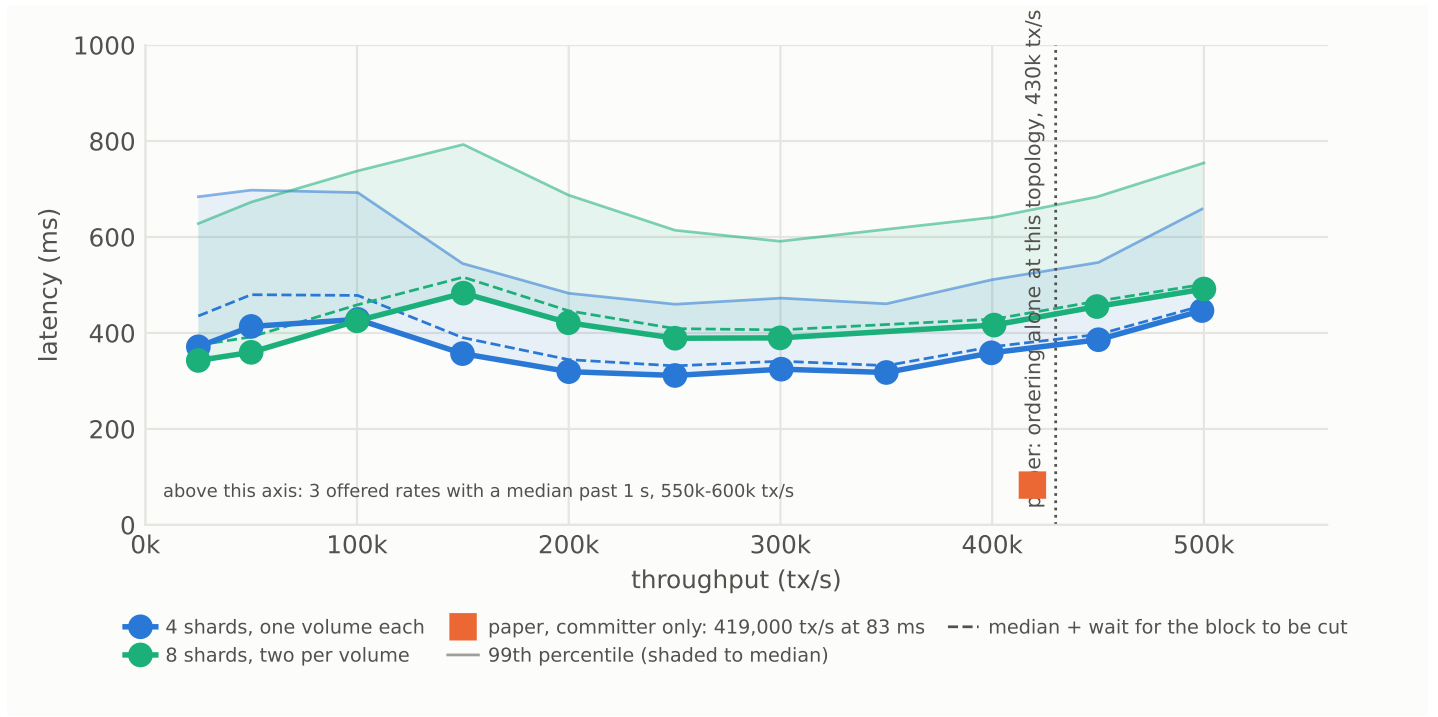


Figure 3: Latency against throughput end to end, with a real ordering service in the path, by shard count and batcher storage layout. Marks as Fig. 2. The dotted vertical line is the SIGMOD’26 ordering-alone figure for this topology.

a range of rates. The eight-shard curve shows exactly that shape: p99 rises to 793 ms at 150,000 tps, then falls back as the rate grows enough to fill a batch before the timer fires — 687, 614 and 591 ms at 200,000, 250,000 and 300,000 tps.

## 8 Latency against throughput, by batch size

Batch size is the latency–throughput knob here, and it trades one for the other almost exactly (Fig. 4), at the same workload again. Cutting the batch from 10,000 transactions to 500 removes most of the latency at every rate the two share — p99 197 ms against 693 ms at 100,000 tps, 286 ms against 460 ms at 250,000, and mean latency between a quarter and a half of what large batches give. That is the predicted effect and its direction confirms the mechanism: a batch a twentieth the size fills twenty times sooner, so the 500 ms `BatchCreationTimeout` stops being what cuts it.

What it costs is the ceiling, and the reason is exact. The block rate tracks the offered rate precisely — 200 blocks a second at 100,000 tps, 400 at 200,000, 500 at 250,000 — and between 500 and 600 blocks a second the arm stops working. 250,000 tps is therefore not a transaction limit at all: it is  $500 \times 500$ , a *block-rate* ceiling near 500 a second. At 10,000 transactions a batch the same block rate would carry five million transactions a second, which is why that configuration is limited by transactions instead and reaches 499,091.

So the two configurations are bounded by different things, and the choice is a real trade rather than a tuning win: small batches halve the latency and halve the ceiling.

## 9 Throughput against transaction size

Throughput falls with transaction size, but far more slowly than in the SIGMOD’26 sweep, and the byte rate is what shows why (Fig. 5). The sweep keeps the same two read–write operations and varies only the value each one carries, so the serialized envelope is 202 bytes plus twice that value. That measurement is bandwidth-bound past 300 B: each of its points sits within 10 % of  $120 \text{ MB s}^{-1}$ , so its rate is simply bytes per second divided by size — and the figure is the cap its own storage carried, since those experiments ran on AWS VMs on EBS General Purpose volumes rated at  $125 \text{ MiB s}^{-1}$ . This deployment is not bandwidth-bound anywhere in the range. Its byte rate *rises* with size, from  $122 \text{ MB s}^{-1}$  at 300 B to  $410 \text{ MB s}^{-1}$  at 4 KiB, because what it pays per transaction dominates what it pays per byte: halving the transaction count at twice the size is a net gain. Throughput therefore falls about  $0.76\times$  per doubling rather than the  $0.5\times$  a bandwidth-bound service gives.

The tabulated 512 B figure is nominally above the 300 B one, which is not a real inversion: the 300 B point’s two holds span 408,000 to 443,800 tps and 512 B sits inside that, so the sweep is flat between those sizes rather than rising. It is the kind of difference the repeatability figure in Table 1 exists to rule out.

Where that stops is measurable, and it is the assembler’s disk. At 4 KiB the arm delivers 113,000 to 123,000 tps whether it is offered 118,000 or 163,200, so throughput plateaus near 120,000 tps —  $492 \text{ MB s}^{-1}$  of envelope bytes — with no machine anywhere above 54 % CPU, so it is not a processing limit. The highest rate that also meets the latency bound, which is what Table 1 reports, is lower still

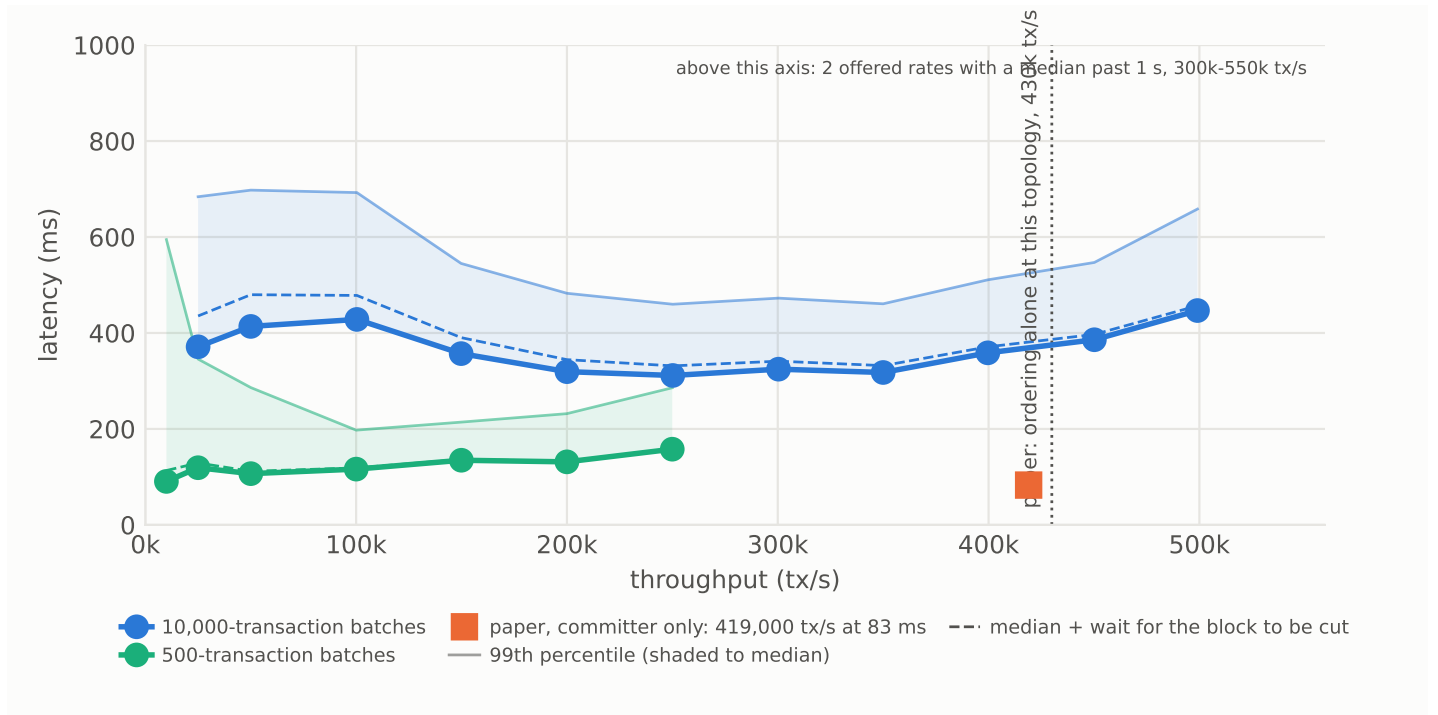


Figure 4: Latency against throughput end to end, by batch size, on four shards with a volume per batcher. Marks as Fig. 2.

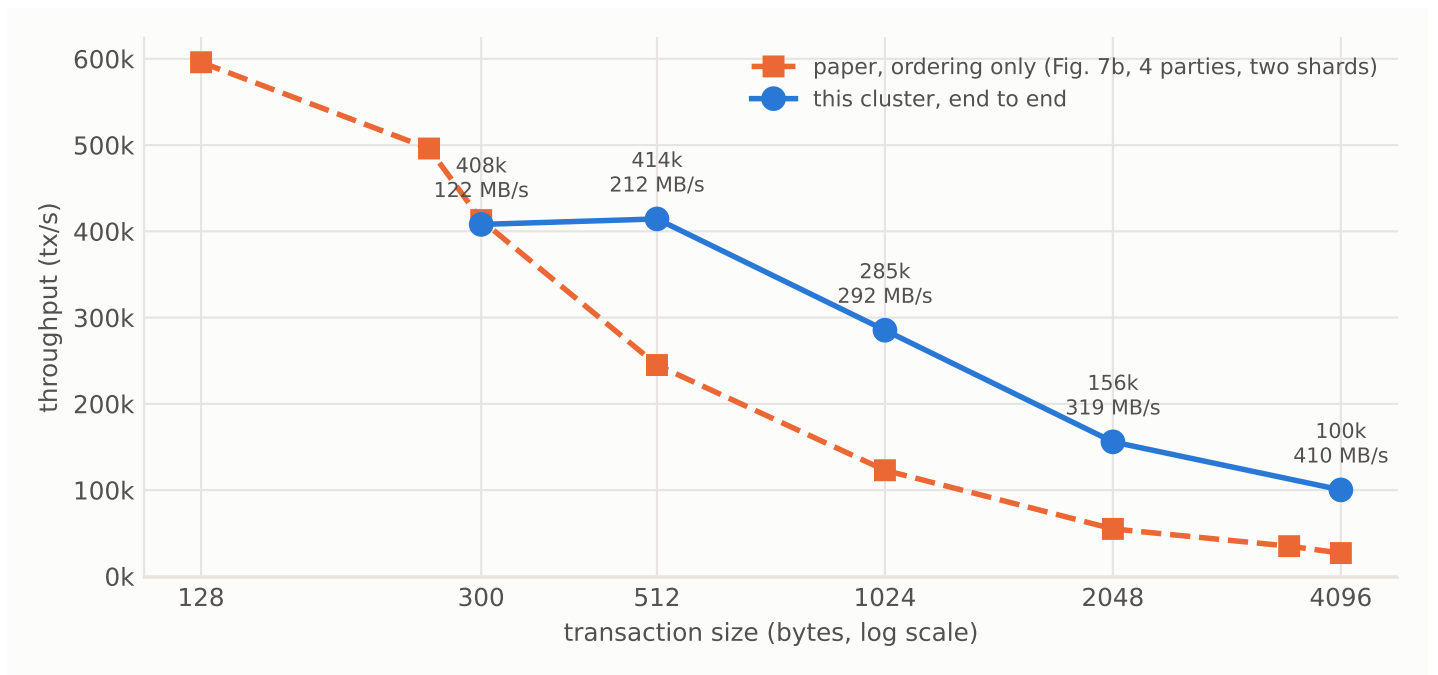


Figure 5: Throughput against transaction size end to end, beside the SIGMOD'26 paper's Figure 7b. Each point is annotated with the byte rate it implies.

at 100,000 tps. The assemblers at that point are writing  $527 \text{ MB s}^{-1}$  to  $546 \text{ MB s}^{-1}$  to their data volume, against the  $529 \text{ MiB/s}$  sequential write Table 2 measures on these machines: the device is saturated. Each of the four assemblers writes the *complete* block stream, where a batcher writes only its own shard's share, so the assembler is where a byte rate first meets a disk. At 2 KiB the same figure is  $319 \text{ MB s}^{-1}$ , three fifths of that ceiling, which is why that point holds cleanly and this one does not.

So the sweep has two regimes rather than one. Below 2 KiB

the arm is limited per transaction and the byte rate climbs; at 4 KiB it is limited per byte by one component's disk, and the SIGMOD'26 bandwidth-bound model finally applies here too, at about four times the bandwidth.

The consequence for the comparison is that only the direction is shared. Both lines fall, and this one is above the SIGMOD'26 line at every size they have in common — but it is a whole pipeline against one of its parts, at four shards against two, so the gap is not a speedup and the shapes are not measuring the same limit.

Table 1: The size sweep end to end, with the evidence class of each point, since they do not all meet the same bar. A hold is tabulated in preference to a higher probe wherever both exist. At 1 KiB and above a hold does not fit in one deployment: each assembler writes the complete block stream, so a 485 GB volume fills in under twenty minutes at these byte rates, less than a rate search followed by a 300 s hold. The 300 B point is the lower of two holds 8.8 % apart, so all figures are given to three significant figures.

| Size  | Throughput (tx/s) | Byte rate | Evidence             |
|-------|-------------------|-----------|----------------------|
| 300 B | 408,000           | 122 MB/s  | two holds, lower one |
| 512 B | 414,000           | 212 MB/s  | confirmed hold       |
| 1 KiB | 285,000           | 292 MB/s  | probe; volume filled |
| 2 KiB | 156,000           | 319 MB/s  | confirmed hold       |
| 4 KiB | 100,000           | 410 MB/s  | probe; disk-bound    |

Table 2: **fio 3.35**, **libaio**, **direct=1**, 20 s per pattern, run on a volume the deployment was not using at the time. Committer figures are the range across a database node and the sidecar. A synchronous engine caps fio’s queue depth at 1 while still reporting the depth asked for, so the deep rows use **libaio**.

| Pattern             | Depth | Committer    | Ordering    |
|---------------------|-------|--------------|-------------|
| 1 MiB seq. write    | 4     | 1,058 MiB/s  | 529 MiB/s   |
| 4 KiB O_DSYNC write | 1     | 2,400 IOPS   | 3,249 IOPS  |
| 8 KiB rand. read    | 32    | 126,000 IOPS | 67,689 IOPS |
| 8 KiB rand. write   | 32    | 102,000 IOPS | 51,165 IOPS |

## 10 Storage characterisation

Two properties of these volumes bound the results above, and both had to be measured: the volumes are **virtio** devices reporting neither a model nor a meaningful rotational flag, so the backing media cannot be identified from inside the guest. Table 2 gives **fio 3.35** in the four patterns this deployment produces.

First, ordering machines have half the disk bandwidth as well as half the cores, at almost exactly a factor of two on every bandwidth row, at round figures repeating to four significant figures across machines and volumes — these are provisioned caps, not device characteristics, so a ceiling on that arm must be read against the cap before it is attributed to ordering. The synchronous pattern is the exception, where the ordering volumes are the faster of the two (297  $\mu$ s against 395  $\mu$ s at the median). Second, durability is the expensive direction: making one 4 KiB record durable with an **O\_DSYNC** write costs about 400  $\mu$ s, capping any per-record design near 2,400 operations a second, three orders of magnitude below the rates measured here. The ledger survives that only by batching — it appends a whole block and syncs every hundredth. Syncing alone is cheap by comparison, an **fdatasync** after a buffered write returning in about 0.08 ms: it is the write, not the barrier, that costs.

## 11 Threats to validity

The load generator is the instrument, and in one figure it is the limit: its block preparation caps the small-block curve in Fig. 2. Elsewhere it is close to being one. At the end-to-end knee it runs at 77 % of its 64 cores against the busiest validator–committer’s 81 %, and on the 300 B size point at 69 % against 75 % — close enough that neither is clearly the

constraint. Every figure near 500,000 tps should therefore be read as a bound on the pair, generator and pipeline together, not on the pipeline alone. Separating them needs a second generator, and there is no spare machine for one without taking it from the committer. A knee is resolved to the search’s 8 % step, so it is a lower bound, drawn one-sided in Fig. 1. The baseline drifts 10 % between days, so only same-day comparisons are used, and within a day repeatability is no better: one configuration measured twice gave holds 8.8 % apart. A difference under about a tenth is therefore not a result here, and no claim in this section rests on one.

A ladder climbs on a single deployment, so rate and table fill rise together along it. Repeating the rates around the knee on fresh deployments moves the boundary by less than the spread above, so the knee is a property of the rate; the tail near it does grow with fill. Every rung records the fill it ran against.